

Synchronization - Part 2

Operating Systems – EDA093/DIT401

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What to read (main textbook)

- Chapter 2.3.7-2.3.9, 2.3.11, 6.3, 2.4.1-2.4.6, 6.1-6.2, 6.5-6.6, 6.7.3-6.7.4

Agenda

- Bounded-buffer producer/consumer
- Resource allocation, deadlocks, and necessary conditions for deadlocks
- Dining philosophers
 - Breaking the circular wait condition
 - Breaking the no-preemption condition
 - Breaking the hold-and-wait condition
- Lamport's bakery algorithm
- Readers/Writers problem
- OS as arbitrator for deadlock avoidance/recovery

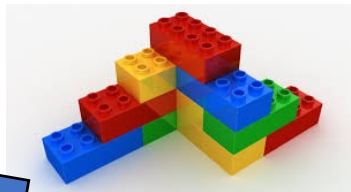
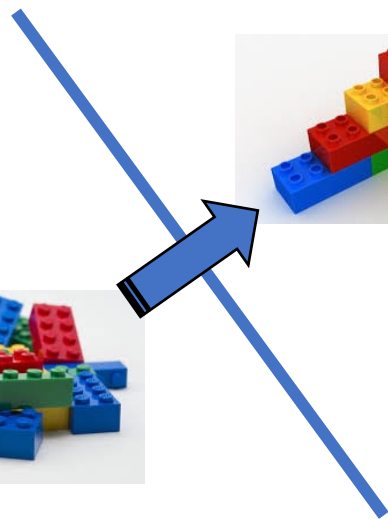
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The Bounded buffer producer-consumer problem

By using primitives ...

- R/W variables
- RMW variables
- Semaphores, etc.
- ...

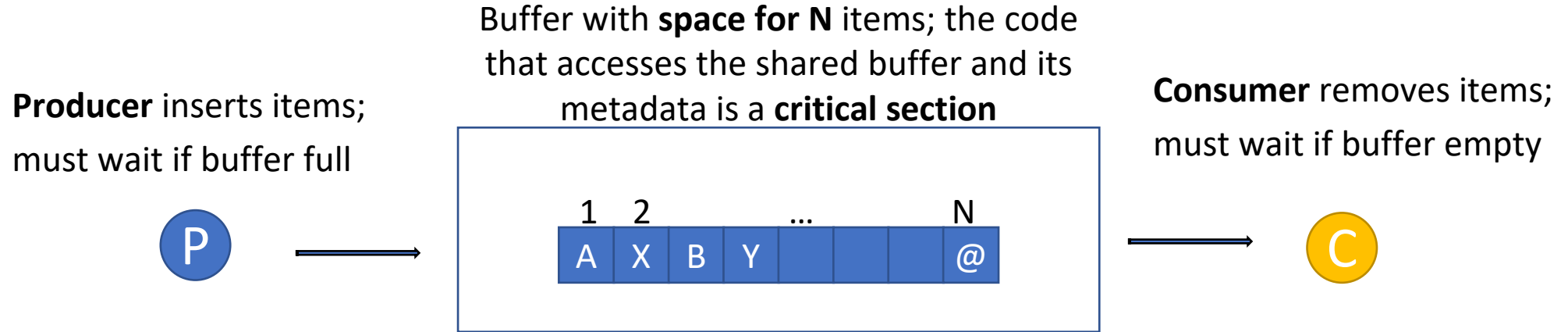


... we want to construct objects / solve specific synchronization problems

- 2-thread CS, n-thread CS
- Producer-consumer (bounded buffer)
- Dining philosophers
- Transactions
- ...

the Bounded buffer producer-consumer problem

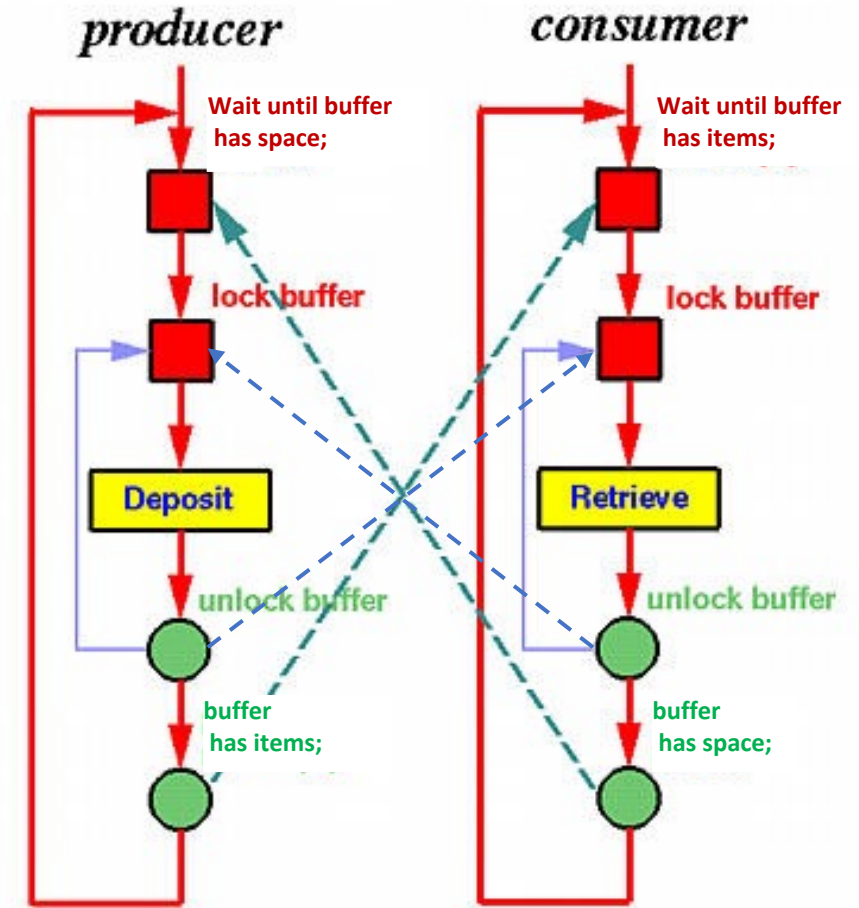
6



Solve this synchronization problem using semaphores

What synchronization do we need?

- **Producer** inserts items; must wait if buffer full
- **Consumer** removes items; must wait if buffer empty
- Accessing the buffer is a **critical section**



a solution to the Bounded buffer producer-consumer problem 8

Synchronization variables:

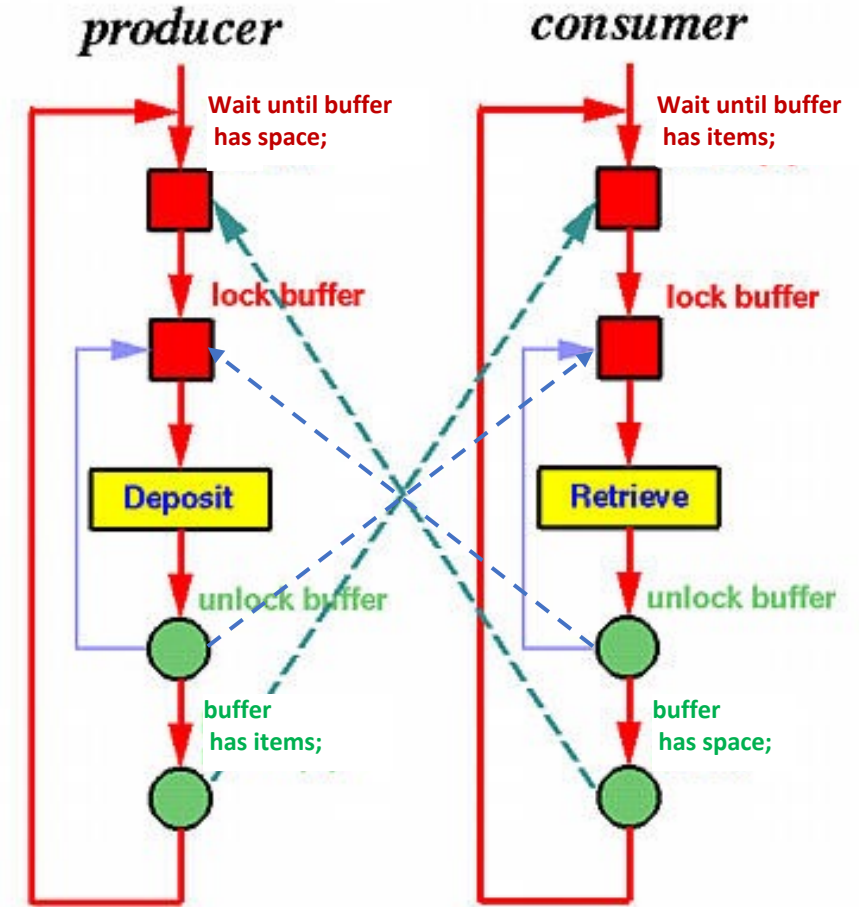
- Binary semaphore **mutex_sem** initialized to 1
- Counting semaphore **buffer-has-items** initialized to 0
- Counting semaphore **buffer-has-space** initialized to N

producer

```
do {  
    // produce item  
    wait(buffer-has-space);  
    wait(mutex_sem);  
    // add item to buffer  
    signal(mutex_sem);  
    signal(buffer-has-items);  
} while (TRUE);
```

consumer

```
do {  
    wait(buffer-has-items);  
    wait(mutex_sem);  
    // remove item from buffer  
    signal(mutex_sem);  
    signal(buffer-has-space);  
    // use the item  
} while (TRUE);
```



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Resource allocation

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- Processes/threads need resources (e.g., memory pages, a printer, access to parts of a shared data structure, etc.)
 - Our focus: reusable resources
- a human analogy: process = go fishing; needed resources: boat, fishing rod
- Mutual exclusion is needed for resources that cannot be shared



Structure of process/thread P

Repeat

Request resources (entry section)

critical section

Release resources (exit section)

remainder section

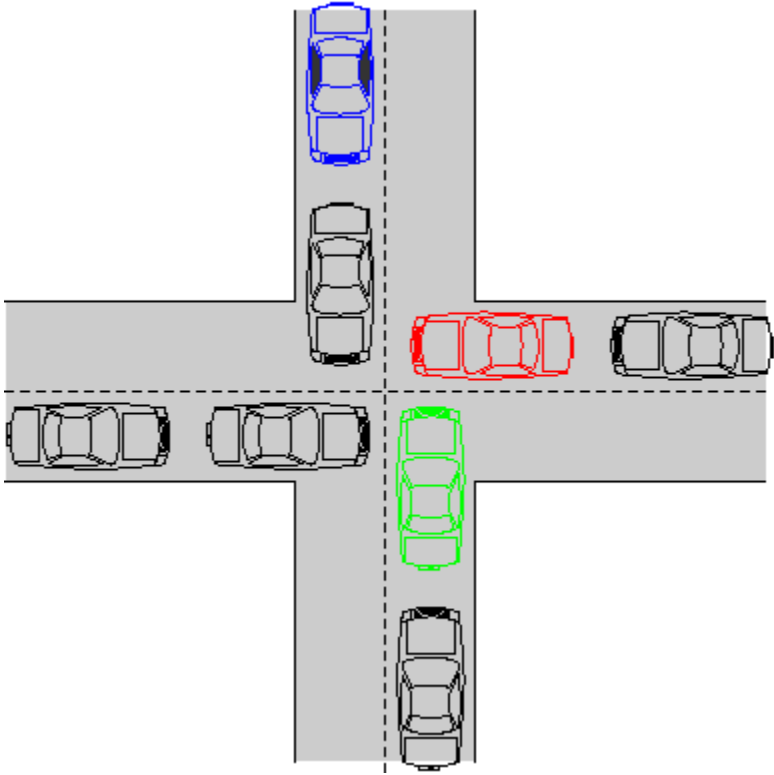
Forever

Problem formulation:

A solution for acquiring/releasing all the needed resources must **provide** the **entry** and **exit** sections **ensuring**:

- 1. Mutual Exclusion.**
- 2. Progress: no deadlock**
- 3. Fairness (e.g., Bounded Waiting, no starvation)**

What is a deadlock?

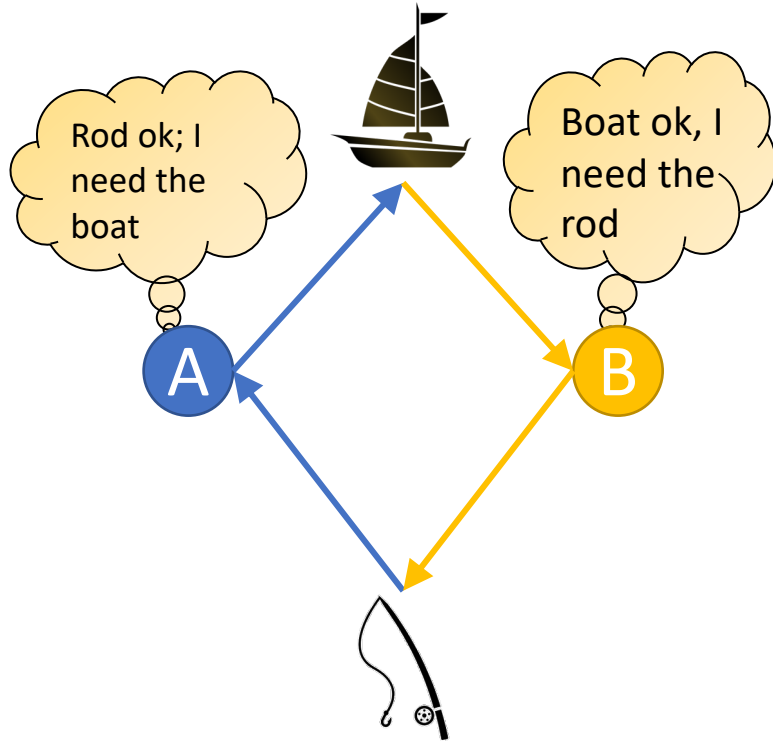


A set of processes/threads blocking each other so that none of them can proceed:

How can it occur?

4 necessary conditions for Deadlock [Coffman et al., 1971]

Theorem: all 4 conditions hold simultaneously when a deadlock occurs:

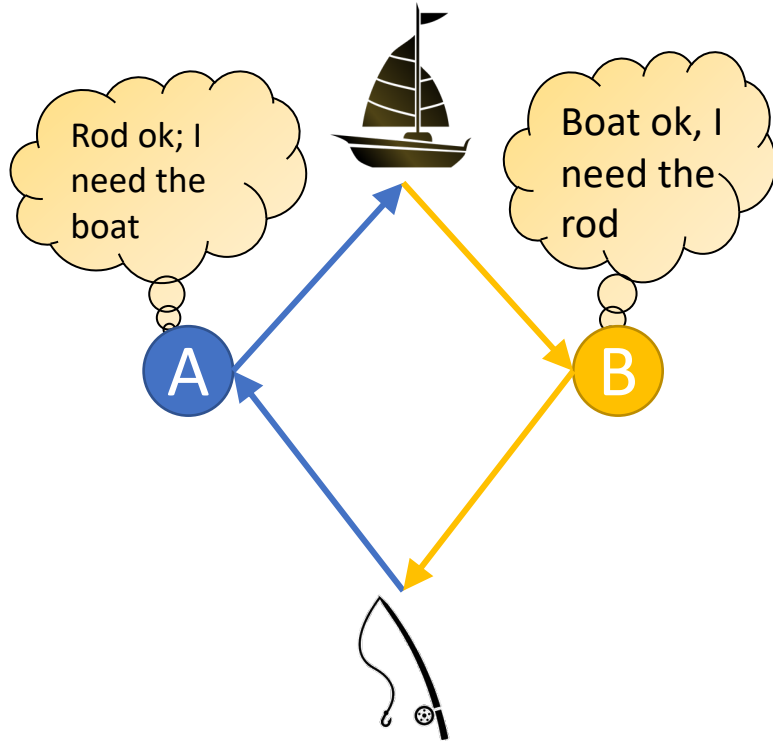


1. **Mutual exclusion:** only one process at a time can use a resource (that is not shareable).
2. **Hold and wait:** a process holding some resource can request additional resources and wait for them if they are held by other processes.
3. **No preemption:** a resource need not be held until the entire task is complete; it is released voluntarily after the process finishes using it.
4. **Circular wait:** there exists a circular chain of 2 or more blocked processes, each waiting for a resource held by the next process in the chain

NOTE: we are talking about preemption of resources, this is not the process scheduling preemption

4 necessary conditions for Deadlock [Coffman et al 1971]

Theorem: all 4 conditions hold simultaneously when a deadlock occurs:



1. **Mutual exclusion:** only one process at a time can use a resource (that is not shareable).
2. **Hold and wait:** a process holding some resource can request additional resources and wait for them if they are held by other processes. → **GET ALL RESOURCES AT ONCE**
3. **No preemption:** a resource can only be released by the process holding it, after that process has completed its task. → **THREADS RELEASE RESOURCES IF THEY MANAGE TO GET SOME BUT NOT ALL**
4. **Circular wait:** there exists a circular chain of 2 or more blocked processes, each waiting for a resource held by the next process in the chain. → **ACQUIRE RESOURCES IN A CERTAIN ORDER**

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Dining philosophers [Dijkstra, 1965]

n philosophers (processes); each philosopher P_i , when hungry, needs both left & right chopsticks, in order to eat

Structure of process/thread P

Repeat

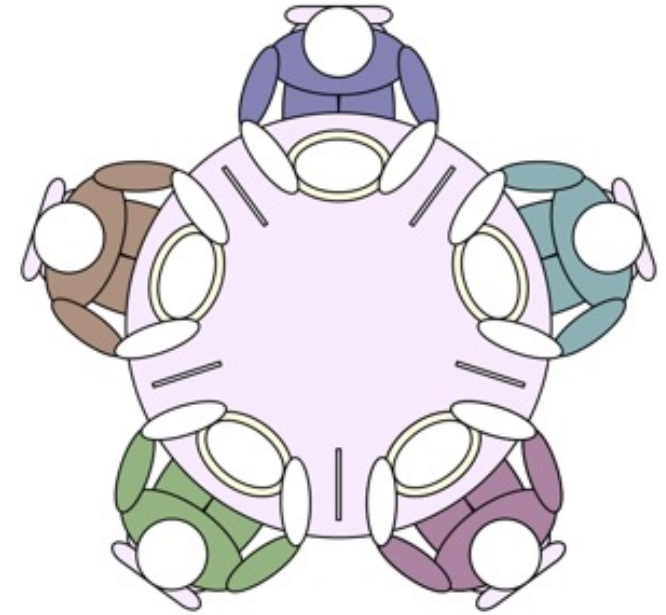
Request resources (left + right chopstick)

eat

Release resources (left + right chopstick)

think

Forever



Dining philosophers: pick-left-then-right approach

```
1.  Shared var c[0..n-1]: binary semaphore // one for
    each chopstick; initially all set to 1
2.  Pi:
3.  do
4.      wait c[i]; // pick left chopstick
5.      wait c[(i+1) mod n]; // pick right chopstick
6.      // Eat
7.      signal c[i]; // leave left chopstick
8.      signal c[(i+1) mod n]; // leave right chopstick
9.      // Think
10. forever
```

Recall the requirements:

1. **Mutual exclusion:** each resource is used by only one process at a time
2. **Progress:** no deadlock
3. **Fairness:** FCFS, or no starvation, or other fairness formulation

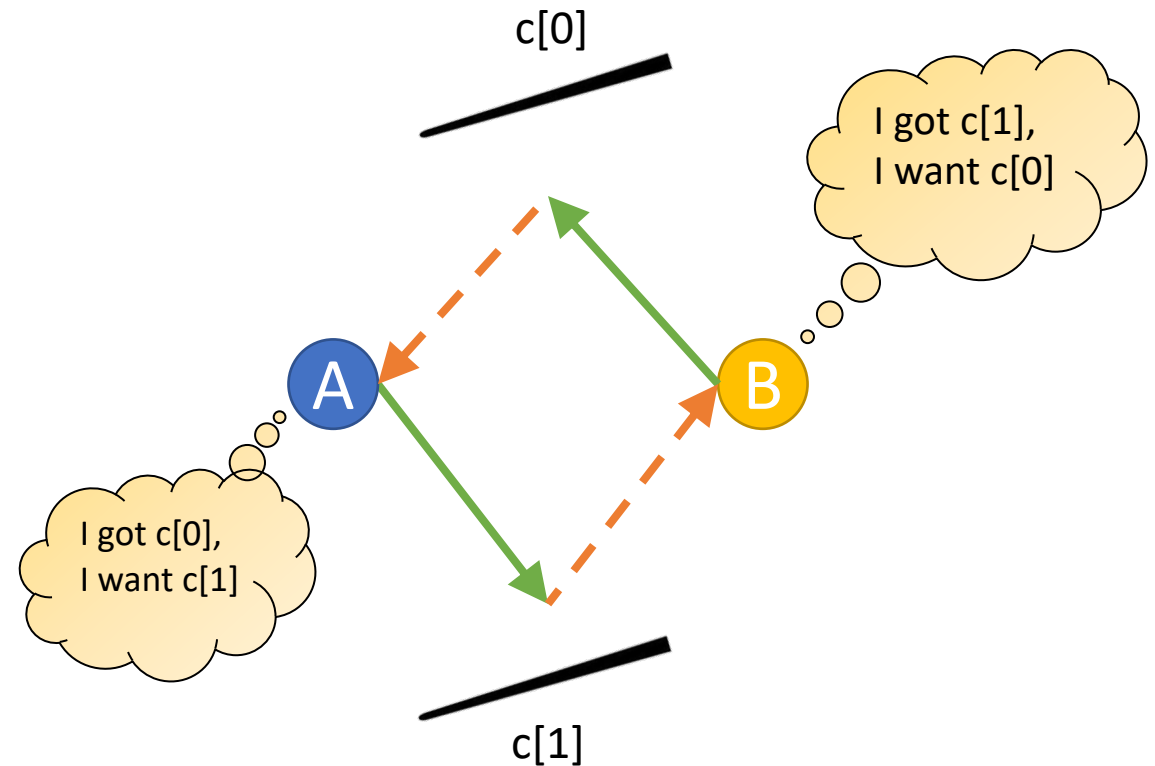
Dining philosophers: pick-left-then-right approach

```
1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2. Pi:
3. do Hold and wait
4.   Wait c[i]; // pick left chopstick
5.   Wait c[(i+1) mod n]; // pick right chopstick
6.   // Eat
7.   Signal c[i]; // leave left chopstick
8.   Signal c[(i+1) mod n]; // leave right chopstick
9.   // Think No preemption
10. forever
```

Dining philosophers: pick-left-then-right approach

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1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2. Pi:
3. do                                     Hold and wait
4.   Wait c[i]; // pick left chopstick
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7.   Signal c[i]; // leave left chopstick
8.   Signal c[(i+1) mod n]; // leave right chopstick
9.   // Think                             No preemption
10. forever
  
```



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Dining philosophers: pick-one-at-a-time without circular wait

```

1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2.  $P_i$  (except  $P_{n-1}$ ):
3.   do
4.     Wait c[i]; // pick left chopstick
5.     Wait c[(i+1) mod n]; // pick right chopstick
6.     // Eat
7.     Signal c[(i+1) mod n]; // leave right chopstick
8.     Signal c[i]; // leave left chopstick
9.     // Think
10. forever

```



```

1. Shared var c[0..n-1]: bin-semaphore // one for
   each chopstick; init all 1
2.  $P_{n-1}$ :
3.   do
4.     Wait c[(i+1) mod n]; // pick right chopstick
5.     Wait c[i]; // pick left chopstick
6.     // Eat
7.     Signal c[i]; // leave left chopstick
8.     Signal c[(i+1) mod n]; // leave right chopstick
9.     // Think
10. forever

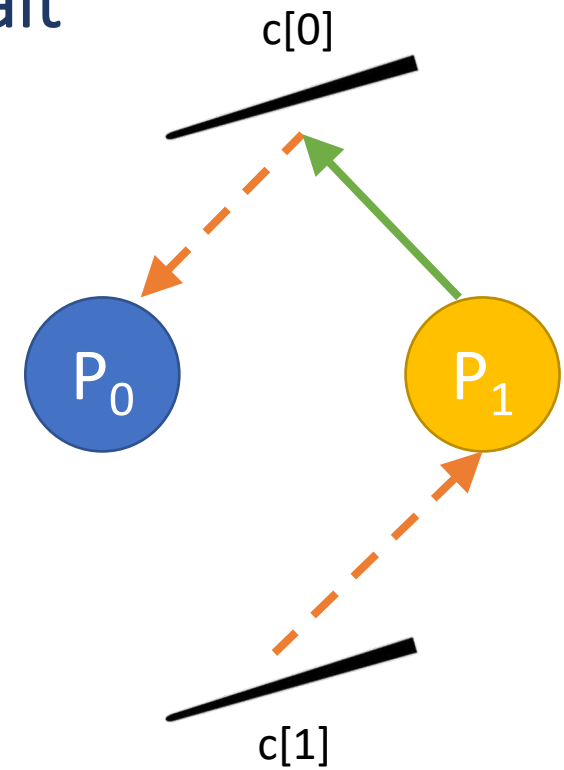
```

Dining philosophers: pick-one-at-a-time without circular wait

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```
1.  Shared var c[0..n-1]: bin-semaphore // one
    for each chopstick; init all 1
2.  Pi (except Pn-1):
3.  do
4.    Wait c[i]; // pick left chopstick
5.    Wait c[(i+1) mod n]; // pick right chopstick
6.    // Eat
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8.    Signal c[i]; // leave left chopstick
9.    // Think
10. forever
```

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1.  Shared var c[0..n-1]: bin-semaphore // one
    for each chopstick; init all 1
2.  Pn-1:
3.  do
4.    Wait c[(i+1) mod n]; // pick right chopstick
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9.    // Think
10. forever
```



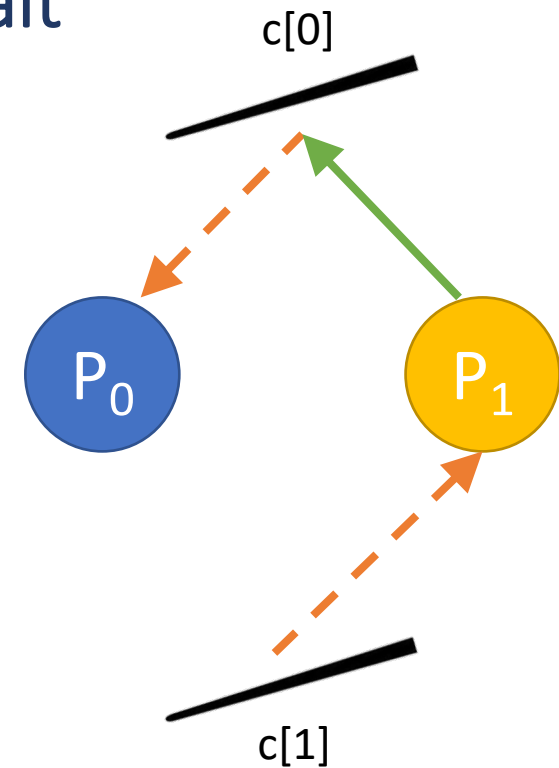
Dining philosophers: pick-one-at-a-time without circular wait

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7.    Signal c[(i+1) mod n]; leave right chopstick
8.    Signal c[i]; // leave left chopstick
9.    // Think
10. forever
  
```

```

1.  Shared var c[0..n-1]: bin-semaphore // one
    for each chopstick; init all 1
2.  Pn-1:
3.  do
4.    Wait c[(i+1) mod n]; // pick right chopstick
5.    Wait c[i]; // pick left chopstick
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```



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Dining philosophers: break the no-preemption condition

```
shared var c[0..n-1]: of type chopstick_struct { // one for each chopstick
  s: bin-semaphore // init 1
  available: boolean //init true
}
```

```
Pi:
  local var holding_both_chopsticks: boolean; // initially false
  repeat
    while (not holding_both_chopsticks) {
      lock(c[i])
      if !trylock(c[(i+1)%n]) then release(c[i])
      else holding_both_chopsticks := true }
    // Eat
    release(c[i])
    release(c[(i+1)%n])
    holding_both_chopsticks := false
    // Think
  forever
```

```
trylock(c: chopstick_struct):
  wait(c.s)
  if c.available then {
    c.available := false
    ret:= true }
  else
    ret:= false;
  signal(c.s)
  return(ret)
```

Idea: when the second resource is not available, release the first one and retry



```
lock(c : chopstick_struct):
  repeat
  until (trylock(c))
```

```
release(c : chopstick_struct):
  wait(c.s)
  c.available := true
  signal(c.s)
```

Dining philosophers: break the no-preemption condition

- Mutual exclusion: ok
- Progress? no deadlock ... but livelock is possible!
- Fairness? a process can starve...

```
Pi:
  local var holding_both_chopsticks: boolean;
  repeat
    while (not holding_both_chopsticks) {
      lock(c[i])
      if !trylock(c[(i+1)%n]) then release(c[i])
      else holding_both_chopsticks := true }
    // Eat
    release(c[i])
    release(c[(i+1)%n])
    holding_both_chopsticks := false
    // Think
  forever
```

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Dining philosophers: Break the hold-and-wait condition

Idea: philosophers agree on who eats next instead of who gets a chopstick



```
shared var semaphore S[0 .. n-1] // init all 0
shared var semaphore mutex // init 1
shared var state[0 .. n-1] in {HUNGRY, THINKING, EATING} // all initialized as THINKING
Pi:
do
  // think
  enterCS(i) // i.e., get both chopsticks
  // eat
  exitCS(i) // i.e., leave chopsticks
forever
```

```
enterCS(i)
  wait(mutex)
  state[i] := HUNGRY
  help(i)
  signal(mutex)
  wait(S[i])
```

```
help(k)
if state[k] == HUNGRY &&
  state[(k-1) mod n] != EATING &&
  state[(k+1) mod n] != EATING
then {state[k] := EATING ; signal(S[k]) }
```

```
exitCS(i)
  wait(mutex)
  state[i] := THINKING
  help((i-1) mod n)
  help((i+1) mod n)
  signal(mutex)
```

Dining philosophers: Break the hold-and-wait condition

- Mutual exclusion: ok
 - Progress: no deadlock
 - Fairness: a process can starve...
-
- If you are interested in a solution with fairness too:
<https://dl.acm.org/doi/pdf/10.1145/62546.62567>

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Lamport's bakery algorithm

Critical section for n threads:

Idea: before entering its critical section, each thread gets a number. Holder of the smallest number enters the critical section.



The decentralized scheme generates numbers in non-decreasing order; e.g., 1, 2, 3, 3, 4, 5, 5, 5, 6 (note: tickets return to zero, and a later round of threads trying to enter the critical section may start again with tickets with value 1)

If threads P_i and P_j choose the same number:

if $i < j$, then P_i goes first; else P_j goes first.

I.e., we use threads' ids to break ties

Lamport's bakery algorithm

```

Shared var choosing: array [0..n - 1] of boolean (init false);
           number: array [0..n - 1] of integer (init 0);

repeat
  choosing[i] := true;
  number[i] := max(number[0], number[1], ..., number [n - 1])+1;
  choosing[i] := false;
  for j := 0 to n - 1 do begin
    while choosing[j] do [nothing]; // spin
    while number[j] ≠ 0 and (number[j],j) < (number[i], i) do [nothing]; // spin
  end;
  // critical section
  number[i] := 0;
  // remainder section
until false;

```

Why does it satisfy the 3 conditions:

Mutual exclusion: (no 2 threads A and B in CS concurrently): Consider the time between A's decision step and A's entry to CS; A decided to move because:

- B had higher number: when B checks , it will wait for A since A has smaller number
- or B was not interested; when B gets interested, it will choose a number > A's number, hence it will wait

Progress: the thread with the smaller (number, thread ID) can proceed

Fairness: If A waits for B and B exits and wants to enter CS again, if A still waits, B will choose a number > A's, number, hence B cannot bypass A

*This is a more **decentralized** method than e.g., Peterson's: no variable is "writable" by all threads*

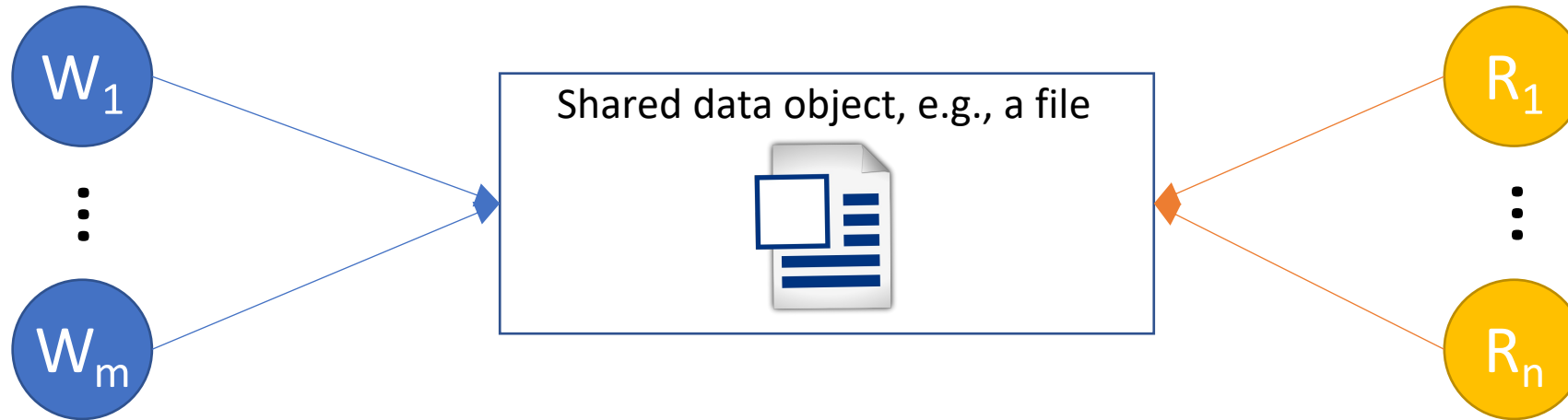
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Readers/Writers problem

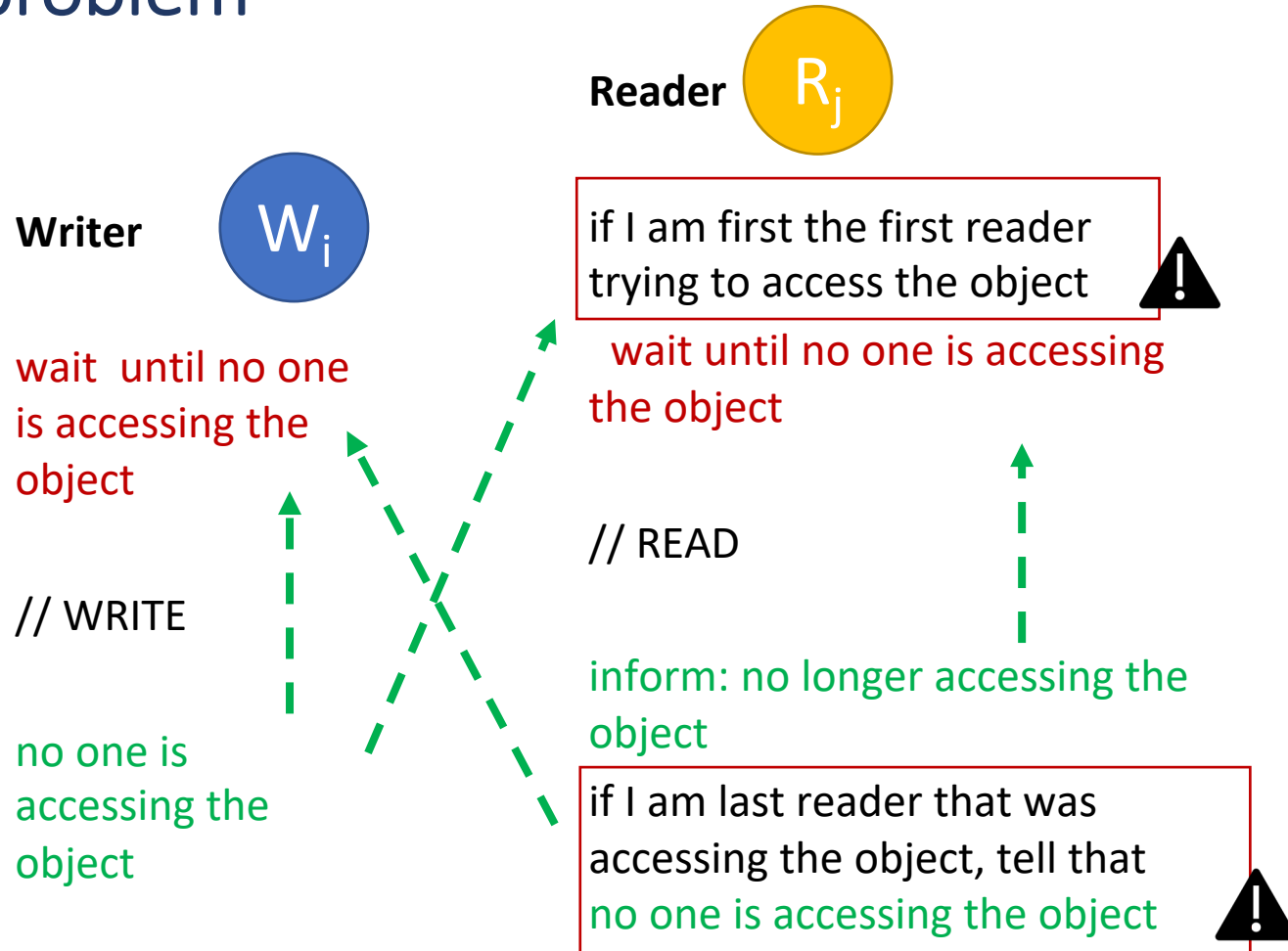
Writers write; must wait if some writer or reader is accessing the shared data object (i.e., *only one writer is allowed to write at a time*)

Readers read; must wait if some writer is accessing the shared data object (i.e., *multiple readers are allowed to read at a time but not concurrently with any writer*)



Solve this synchronization problem using semaphores, ensuring the presented safety properties and progress

Readers/Writers problem



Readers/Writers problem

shared var:

noone_accesses, protect_check: binary semaphore; // initially 1
rc: int ; // active readers counter, init 0



Writer



Repeat

```
wait(noone_accesses);
// WRITE
signal(noone_accesses)
forever
```

Reader



repeat

```
wait(protect_check); // CS to change and check rc variable
rc++;
if rc == 1 then wait(noone_accesses) // "first" reader: block writers or wait
if some of them writes
signal(protect_check);
// READ
wait(protect_check); // CS to change and check rc
rc--;
if rc == 0 then signal(noone_accesses) fi // "last" reader: signals
signal(protect_check)
forever
```

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Deadlock avoidance using the OS as arbitrator

Resource request & allocation is managed by the OS

Deadlock *avoidance*:

- deadlock *might be* possible if resources are granted arbitrarily,
- but OS uses extra info to grant requests and schedule processes so that it avoids deadlock
- **Banker's algorithm [Dijkstra, 1965]** can be used to decide if a resource request can be granted safely

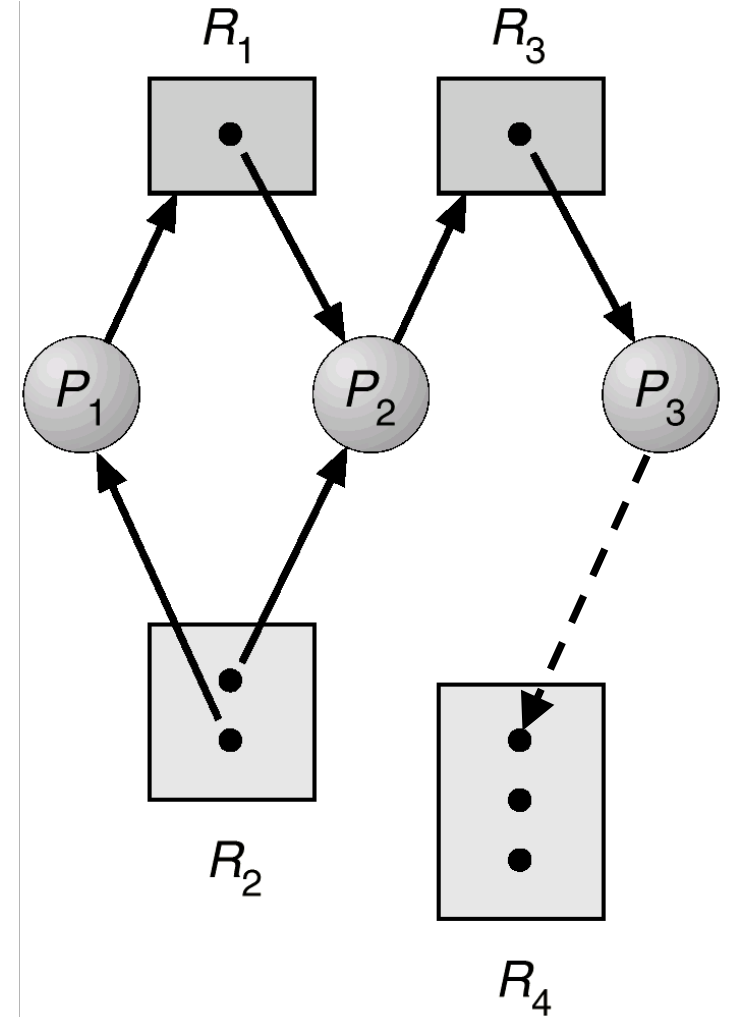
Deadlock avoidance by the OS: system model

Resource types $R_1, R_2, \dots, R_m$

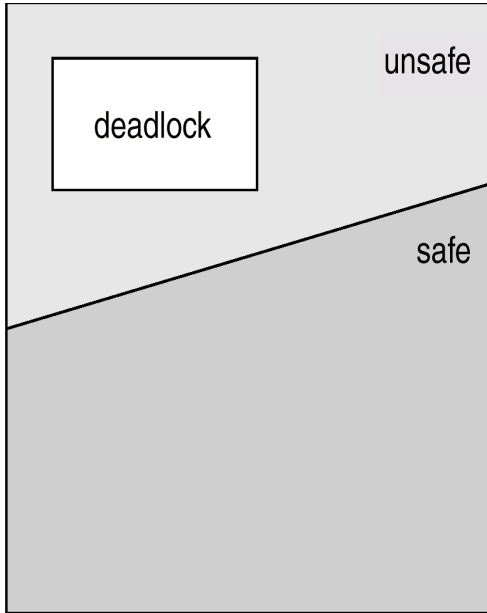
- e.g., I/O devices, files
- each resource type R_i has W_i instances.

Resource-Allocation Bipartite Graph $G(V,E)$

- nodes:
 - $P = \{P_1, P_2, \dots, P_n\}$ the set of **processes**
 - $R = \{R_1, R_2, \dots, R_m\}$ the set of **resource types**
- edges:
 - **claim edge**: $P_i \rightarrow R_j$
 - If dotted: potential request
 - **assignment edge**: $R_j \rightarrow P_i$



Deadlock avoidance using the OS as arbitrator

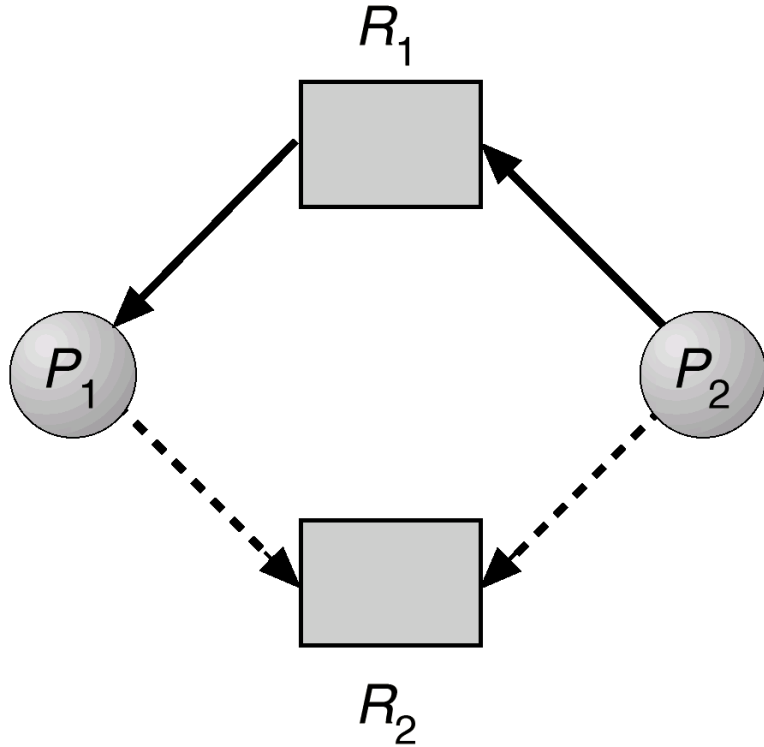


Deadlock-avoidance algo, run by OS:

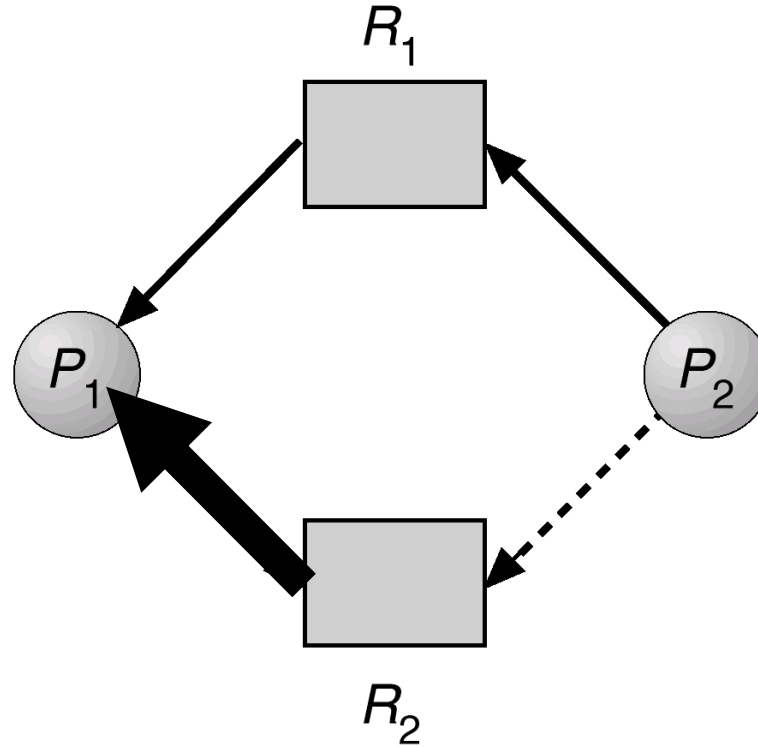
- examines the **resource-allocation state...**
 - *Available, allocated resources*
 - *maximum possible demands* of the processes.
- ...to **ensure there is no potential deadlock:**
 - **unsafe** state $\Rightarrow$ **deadlock might** occur (i.e., later, if all procs request their maximum and no-one can be granted)
- **Avoidance** = ensure that system will not enter an *unsafe* state, by **suspending processes with risky requests**, until enough resources are freed.



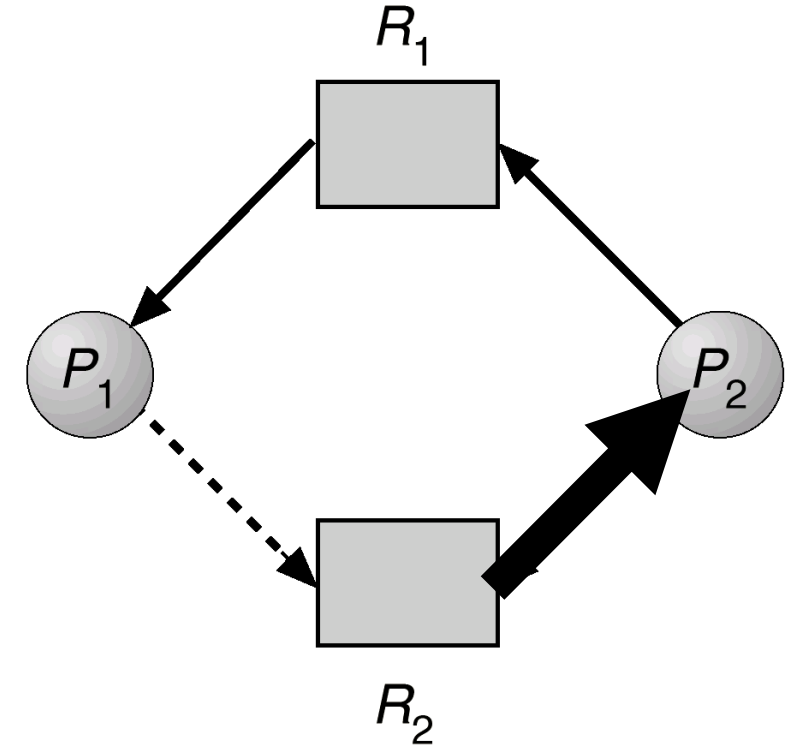
Deadlock avoidance by the OS: example



Example Safe State



Safe State



Example Unsafe State:
i.e., granting the P2-R2 request should not be made until P1 finishes

Deadlock avoidance by the OS: banker's algorithm



Max [i,j] = k:

P_i may request max k instances of R_j .

Allocation[i,j] = k:

P_i holds k instances of R_j

Available [j] = k:

k instances of R_j are still available.

Avoidance = ensure that system will not enter an unsafe state.

Idea:

If *potentially satisfying a request can* result in an *unsafe state* // i.e., bank will have not enough to let its customers finish and return their loans in case someone requests its max needs

then the *requesting process is suspended*

until enough resources are free-ed // by processes that will terminate in the meanwhile

How to do the safety check efficiently?

Banker's algo gives criterion that can be checked in linear time using the Max, Allocation, Available matrices

Deadlock detection and recovery

- If OS grants requests without checking safety upon every request
- It can allow a deadlock state and when detected (e.g., through detection of cyclical waits), *recover*

(1) Process Termination: Abort all or some deadlocked processes until deadlock is eliminated.

(2) Resource Preemption: Select victim and rollback – return to some safe state, restart process from that state

Must decide on selection criteria (cost, starvation risks, ...)

Recovery is pretty expensive as a method